

Case Study

Assessment week 1 | case study 1

Retail Sales & Customer Insights Dashboard-Cloud That

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(Batch 3)



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GitHub: https://github.com/amankumar1ext-web/Case_Study/tree/main/W1_Case_Study_1

CHAPTER 1: INTRODUCTION

1.1 Introduction

This project focuses on analyzing e-commerce order fulfillment and delivery operations using SAP business data. Python was used for data extraction and transformation, while Power BI was used to create interactive dashboards and generate business insights.

1.2 Problem Statement

Organizations need visibility into order processing, delivery performance, carrier efficiency, and customer fulfillment. Lack of centralized reporting makes it difficult to monitor operational performance and identify bottlenecks.

1.2 Objectives

- Analyze order fulfillment performance.
- Monitor delivery and shipment activities.
- Measure customer fulfillment metrics.
- Evaluate carrier performance.
- Develop an interactive Power BI dashboard.

1.4 Scope

The project covers customer, order, delivery, shipment, and carrier data. Analysis is limited to operational and fulfillment-related KPIs available within the dataset.

1.5 Solution Approach

- Extract SAP datasets using Python.
- Perform data validation and cleaning.
- Transform data into analysis-ready tables.
- Build data model in Power BI.
- Create KPIs and dashboards for reporting.

Chapter 2: ETL using Python

2.1 Data Extraction

Extracted SAP data from sheet and rename all sheet as:

SAP TABLE	MEANING	RENAME TO
KNA1	Customer Master	Customers
LFA1	Carrier/Vendor Master	Carriers
VBAP	Order Header	Orders
VBAP	Order Items	OrderItems
LIKP	Delivery Header	Deliveries
LIPS	Delivery Items	DeliveryItems
VTTK	Shipment Header	Shipments
VTTP	Shipment Items	ShipmentItems

```
file_path = "./SAP-DataSet.xlsx"

excel_file = pd.ExcelFile(file_path)

print(excel_file.sheet_names)

['KNA1', 'LFA1', 'VBAP', 'VBAP', 'LIKP', 'LIPS', 'VTTK', 'VTTP']
```

```
Customers = pd.read_excel(excel_file,sheet_name="KNA1")
Carriers = pd.read_excel(excel_file,sheet_name="LFA1")
Orders = pd.read_excel(excel_file,sheet_name="VBAP")
OrderItems = pd.read_excel(excel_file,sheet_name="VBAP")
Deliveries = pd.read_excel(excel_file,sheet_name="LIKP")
DeliveryItems = pd.read_excel(excel_file,sheet_name="LIPS")
Shipments = pd.read_excel(excel_file,sheet_name="VTTK")
ShipmentItems = pd.read_excel(excel_file,sheet_name="VTTP")
```

2.2 Data Validation

Data quality checks were performed to identify missing values, duplicate records, and incorrect data types.

```
"DeliveryItems":DeliveryItems,  
"Shipments":Shipments,  
"ShipmentItems":ShipmentItems  
}.items():  
  
print("\n",name)  
print(df.isnull().sum())
```

```
Customers  
Customer_ID      0  
Customer_Name    0  
Country          0  
Region          0  
City            0  
Postal_Code      0  
Street Address   0
```

2.3 Data Transformation

Column names were standardized, date fields were converted to date format, and required business attributes were prepared for reporting.

```
for name,df in {  
    "Customers":Customers,  
    "Orders":Orders,  
    "OrderItems":OrderItems,  
    "Deliveries":Deliveries,  
    "DeliveryItems":DeliveryItems,  
    "Shipments":Shipments,  
    "ShipmentItems":ShipmentItems  
}.items():  
  
    print(name,df.duplicated().sum())
```

```
Customers 0  
Orders 0  
OrderItems 0  
Deliveries 0  
DeliveryItems 0  
Shipments 0  
ShipmentItems 0
```

```

)

Shipments['Shipment_Date'] = pd.to_datetime(
    Shipments['Shipment_Date']
)

ShipmentItems['Shipment_Date'] = pd.to_datetime(
    ShipmentItems['Shipment_Date']
)

```

```
OrderItems.head()
```

	Sales_Document	Item_Number	Material_Number	Quantity	Net_Price	Item_Status	Delivery_Date
0	1000001	10	MAT001	100	50	Open	2025-02-08
1	1000001	20	MAT002	200	30	Open	2025-02-08
2	1000001	10	MAT003	150	40	Delivered	2025-02-08

2.4 Data Export

The cleaned datasets were exported into CSV files and imported into Power BI for dashboard development.

```

: Customers.to_csv("Customers.csv",index=False)

Carriers.to_csv("Carriers.csv",index=False)

Orders.to_csv("Orders.csv",index=False)

OrderItems.to_csv("OrderItems.csv",index=False)

Deliveries.to_csv("Deliveries.csv",index=False)

DeliveryItems.to_csv("DeliveryItems.csv",index=False)

Shipments.to_csv("Shipments.csv",index=False)

ShipmentItems.to_csv("ShipmentItems.csv",index=False)

```

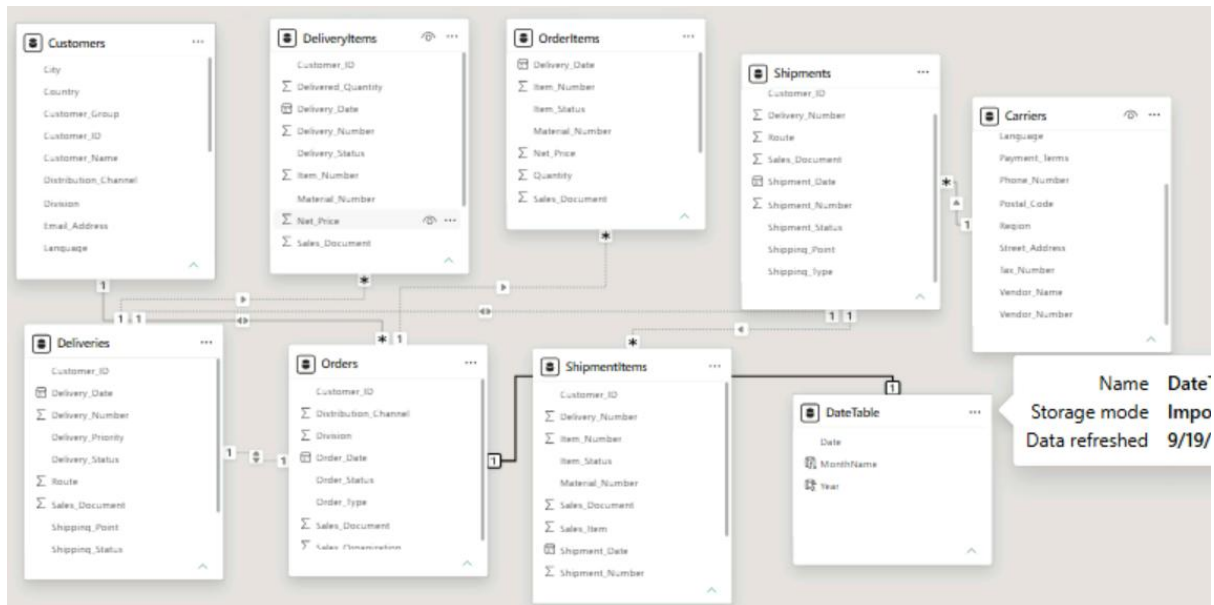
2.5 Chapter Conclusion

The SAP datasets were successfully extracted, validated, transformed, and prepared for analytical reporting.

Chapter 3: Power BI Modeling and Dashboard Development

3.1 Data Modeling

Relationships were established between customer, order, delivery, shipment, and carrier tables to enable cross-functional analysis.



3.2 KPI Development

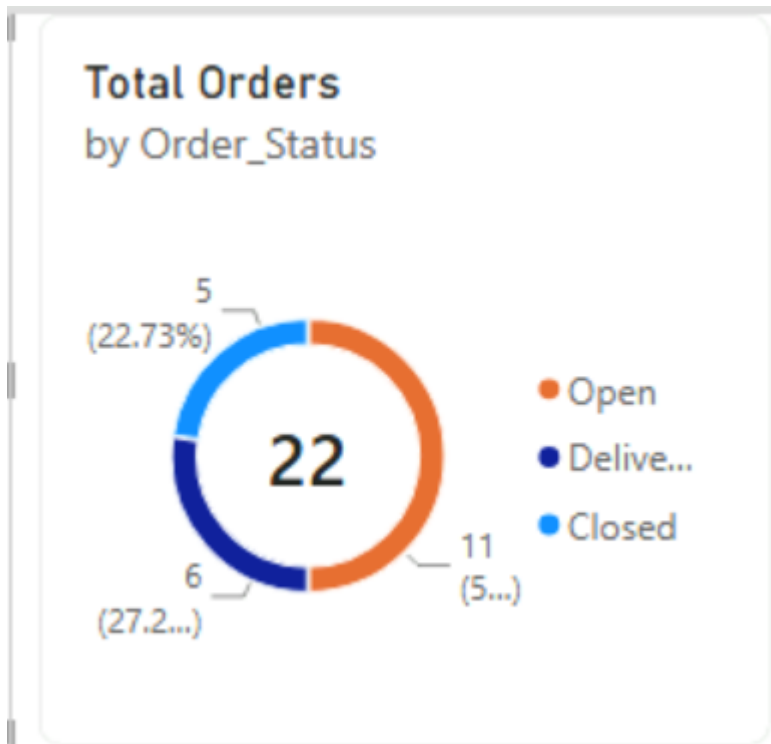
Total Revenue

Measures total revenue generated from all orders.



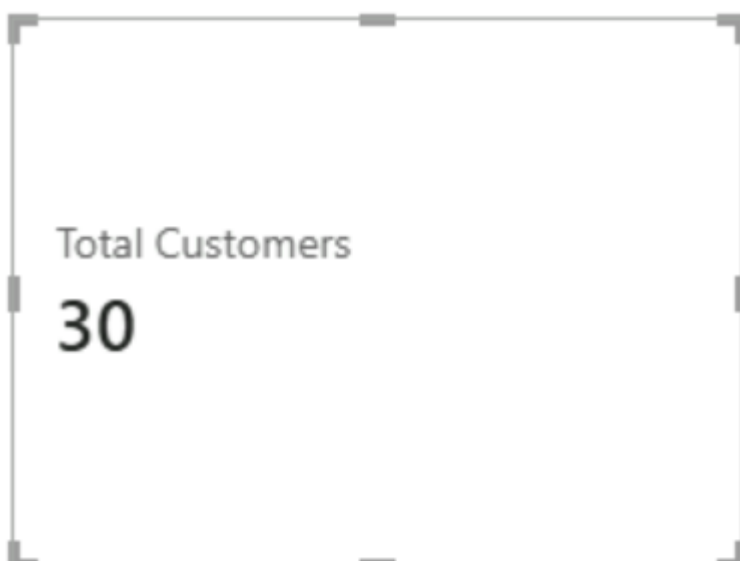
Total Orders

Displays the total number of orders processed.



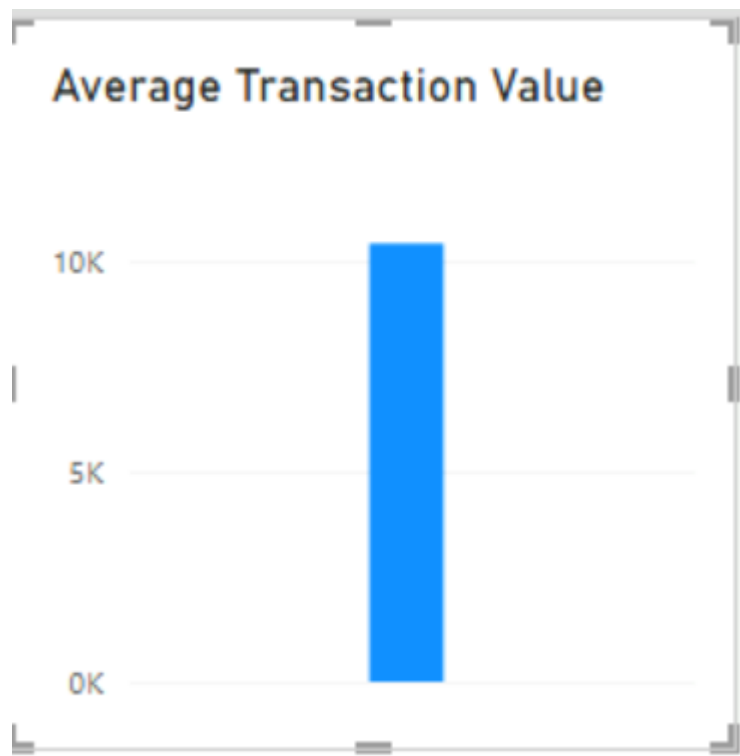
Total Customers

Shows the total number of customers in the dataset.



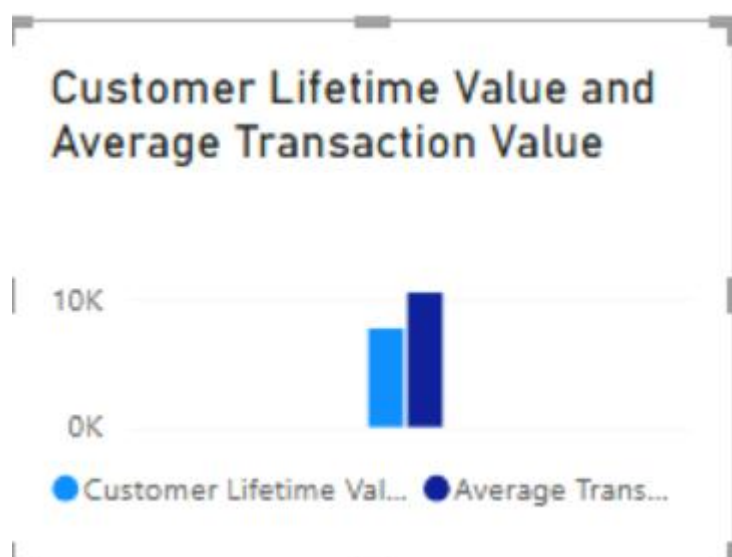
Average Transaction Value (ATV)

Represents the average revenue generated per order.



Customer Lifetime Value (CLV)

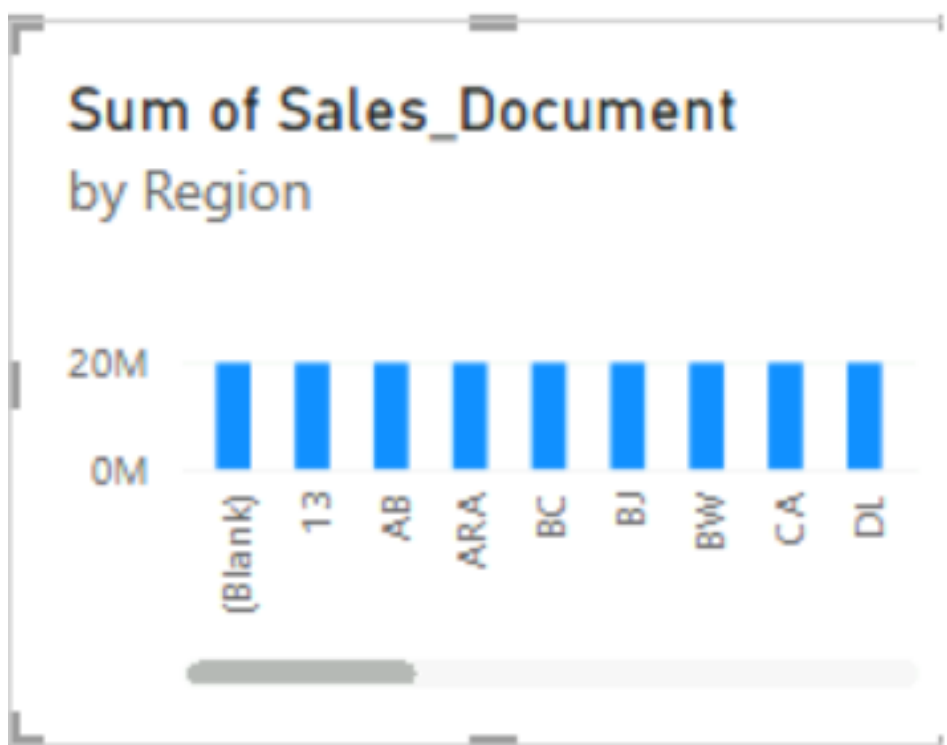
Measures average revenue contribution per customer.



3.3 Sales Analysis

Sales by Region

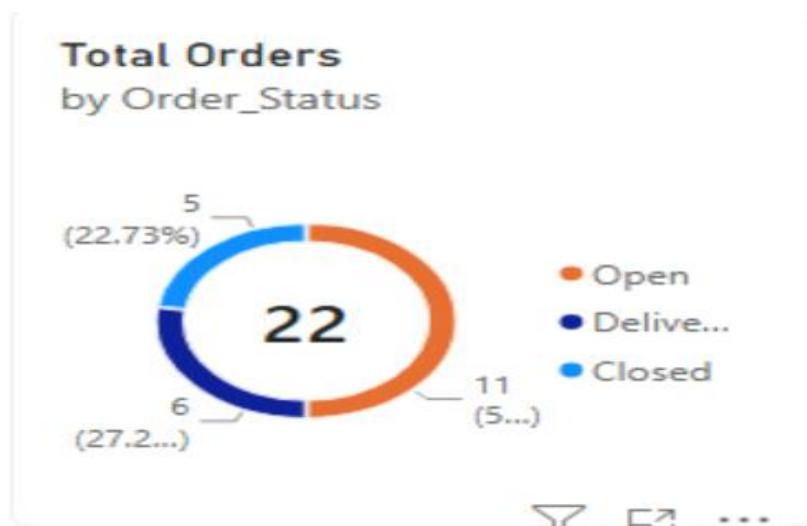
Analyzes revenue generated across different regions.



3.4 Operational Analysis

Orders by Status

Shows distribution of orders across different statuses.



Delivery Performance

Monitors delivery completion status.



Shipment Status Analysis

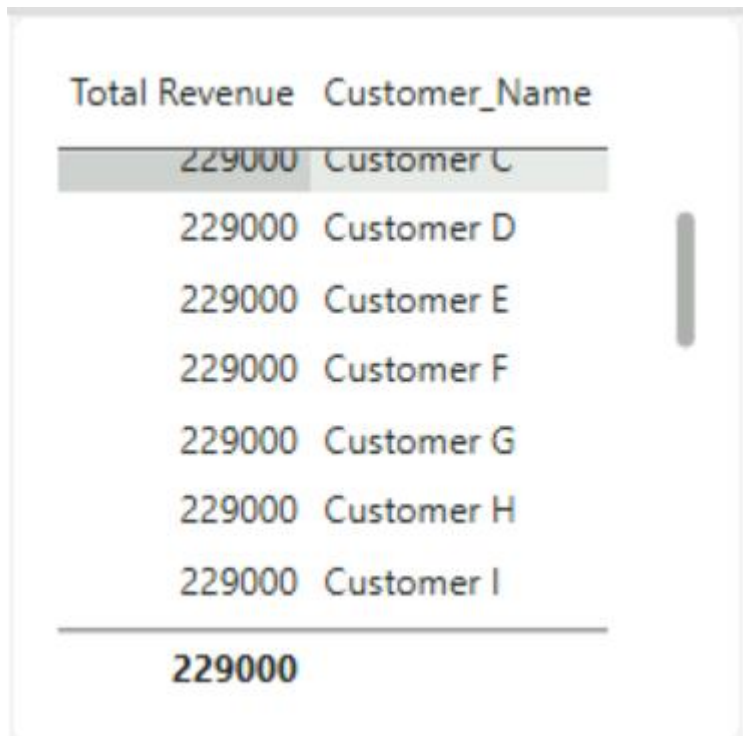
Tracks shipment progress and completion.



3.5 Customer Insights

Top Customers

Identifies customers generating the highest revenue.

A screenshot of a data table with two columns: 'Total Revenue' and 'Customer_Name'. The table lists seven customers, each with a revenue of 229000. The first row, 'Customer C', is highlighted. A vertical scrollbar is visible on the right side of the table. At the bottom, a horizontal line separates the individual rows from a total revenue of 229000.

Total Revenue	Customer_Name
229000	Customer C
229000	Customer D
229000	Customer E
229000	Customer F
229000	Customer G
229000	Customer H
229000	Customer I
229000	

3.6 Key Insights

- Revenue was concentrated among a limited set of customers.
- Certain regions contributed more significantly to sales.
- Fulfillment and delivery performance can be monitored through operational KPIs.
- Carrier performance analysis helps identify logistics efficiency.
- Interactive dashboards enable real-time business monitoring.

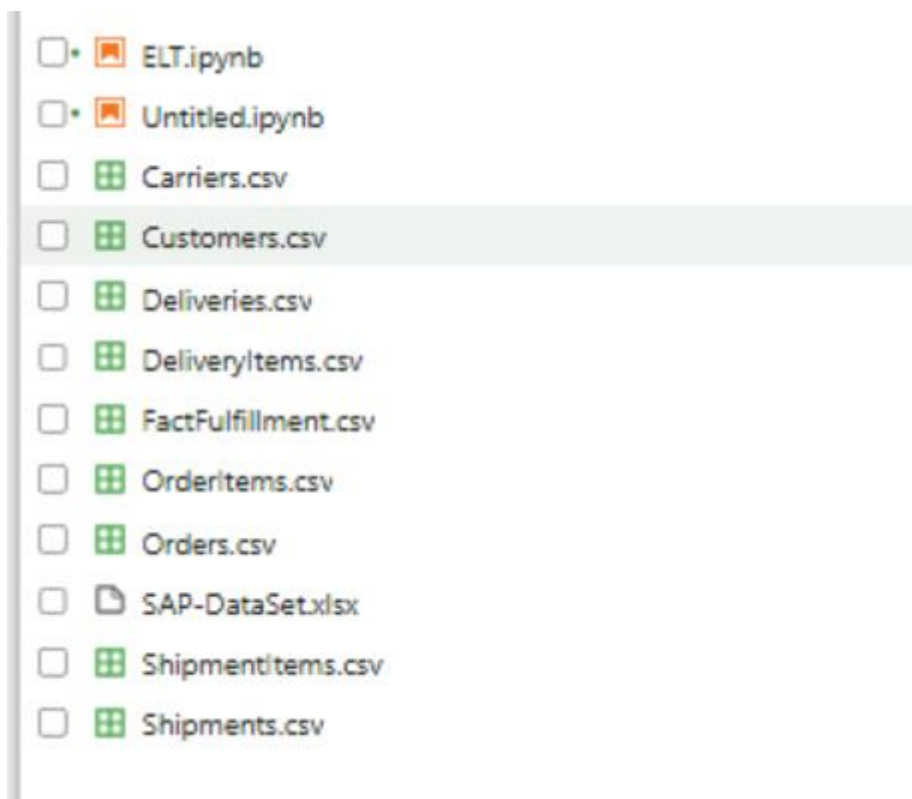
3.7 Conclusion

Power BI dashboards were developed to analyze order fulfillment, delivery operations, customer activity, and sales performance. The solution provides actionable insights for operational and business decision-making.

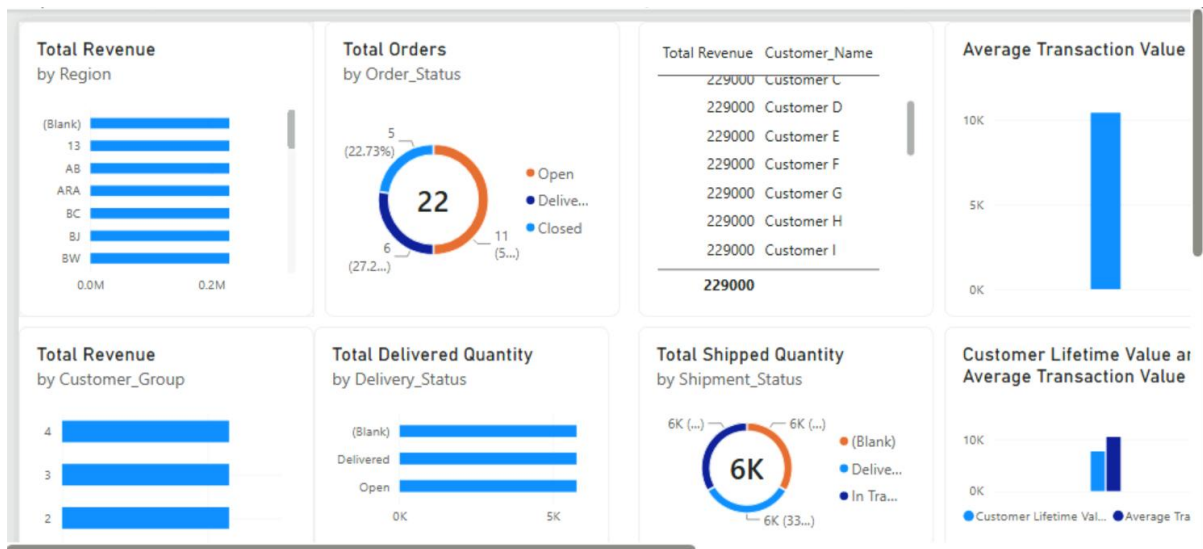
3.7 Conclusion

Power BI dashboards were developed to analyze order fulfillment, delivery operations, customer activity, and sales performance. The solution provides actionable insights for operational and business decision-making.

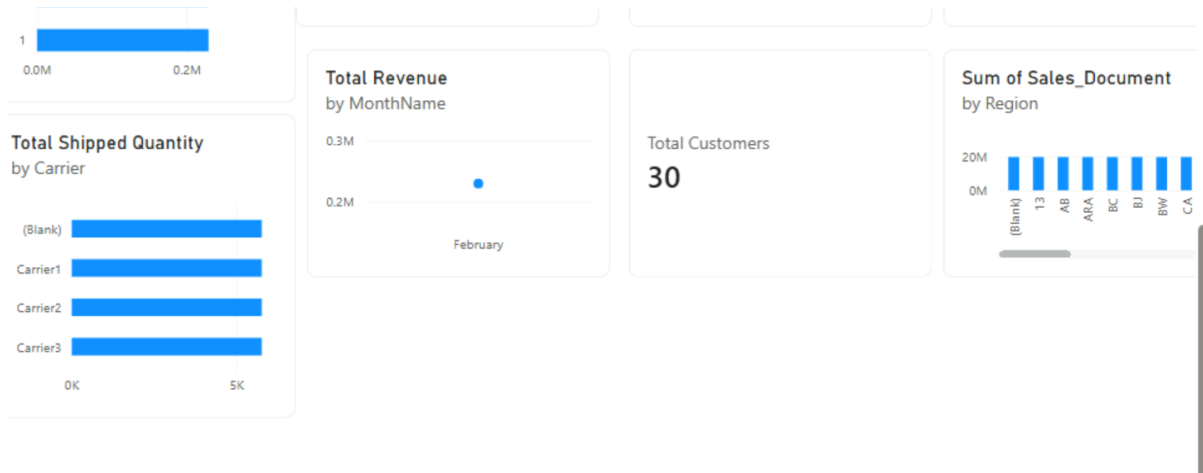
GitHub:https://github.com/amankumarl ext-web/Case_Study/tree/main/W1_Case_Study_1



Transformed CSV files



Dashboards



EOD